AI Usage Log (Annexure A format)

Maintained as the work happens. Tool: Claude Code (Claude Fable 5.1) unless stated otherwise. Decision column: Accept / Modify / Reject, as decided by the team. Entries marked "team to confirm" were proposed by the AI and need a team member's sign-off.

No.	Date	Tool	Purpose / Prompt summary	Output summary	Decision	Reason and verification
1	2026-10-30	Claude Code	Read the course PDF, charter and readme and summarise what must be built	Correct list of deliverables and marking weights; noticed the charter PDF is truncated after "Intended users"	Accept	Checked against the PDFs; truncation confirmed by inspecting the 1-page file
2	2026-10-30	Claude Code	Asked whether the project instructions require an "agent"	Reported the word does not appear in any document; AI's role for Topic 2 is model structuring, formula checking and memo drafting	Accept	grep over all three documents returned no match
3	2026-10-30	Claude Code	Propose base case numbers, time step, platform and sensitivity drivers	Streamlit, monthly 36-month case, ₹20 cr land, 1 lakh sq ft, ₹3,000/sq ft cost, ₹7,500/sq ft price, 60% construction debt at 12%, 25% tax, 15% discount	Modify	Team accepted platform and case; debt definition later changed (entry 7)
4	2026-10-30	Claude Code	Write cash flow engine (land, approvals, S-curve construction, sales, booking-plus-instalment collections, debt sweep, tax)	feasibility/model.py; smoothstep S-curve $3x^2-2x^3$ chosen so it is reproducible in Excel	Accept (team to confirm)	Identity tests: spend totals, collections = revenue, loan fully repaid, equity = project + debt flows
5	2026-10-30	Claude Code	Write NPV, IRR, payback, profit on cost and break-even solver	feasibility/metrics.py	Modify	IRR was WRONG on first run: returned None for the base case. See entry 6
6	2026-10-30	Claude Code	Diagnose IRR = None	AI explained its own bisection bracket (−99% to +1000% per month) had the same NPV sign at both ends because of the final negative tax flow	Accept	Replaced with grid scan plus bisection; test asserts IRR = discount rate at the break-even price; Excel IRR() agrees to 1e−12
7	2026-10-30	Claude Code	Review base-case output	Peak debt only ₹0.79 cr and interest ₹0.04 cr because debt funded construction only and collections repaid it immediately	Modify	Economically unrealistic for a developer who borrows against land; changed loan to fund 50% of land, approvals and construction. Peak debt ₹12.1 cr, interest ₹2.0 cr
8	2026-10-30	Claude Code	Write sensitivity, tornado and scenario module	feasibility/sensitivity.py with 8 drivers, ±10%/±3 months/±2 pp swings, 4 preset scenarios	Accept (team to confirm)	Spot-check: price −10% reduces NPV by roughly revenue × 10% discounted, as expected
9	2026-10-30	Claude Code	Write investment memo generator	Template memo (offline) plus optional Claude memo restricted to a fact sheet, with a check that flags any number not in the fact sheet	Accept (team to confirm)	Template uses only computed values; LLM path not yet exercised (no API key on the build machine)
10	2026-10-30	Claude Code	Build Streamlit app	app.py with KPI rows and six tabs	Modify	First run crashed the cash flow table (mixed integer and "Total" index); fixed. Deprecated width argument replaced
11	2026-10-30	Claude Code	Generate Excel benchmark with formulas only	benchmark/build_benchmark.py → base_case_benchmark.xlsx; SUMPRODUCT used for lagged collections so instalment months stays adjustable	Accept (team to confirm)	Recalculated in LibreOffice; all 17 metrics match Python within 1e−11 (docs/validation_results.md)
12	2026-10-30	Claude Code	Automate Python vs Excel comparison	benchmark/compare.py	Modify	First version wrote to /tmp which snap LibreOffice cannot access; folder moved into project

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13	2026-10-30	Claude Code	Draft the 3-5 page user manual as a shared Claude Doc (11 sections: purpose, install, screen, inputs, results, cash flow, sensitivity, memo, validation, assumptions, limitations)	Manual drafted with base-case figures taken from the running app	Modify (team to confirm)	AI first wrote that land cost was the third-widest tornado bar; a check of the model showed sales launch delay ranks third. Corrected in the doc. Team to proofread and export to PDF
14	2026-10-30	Claude Code	Write property tests over 200 random input sets and compare NPV/IRR with numpy_financial	tests/test_properties.py; 383 tests total	Modify	AI's own monotonicity assertion "NPV always falls when the discount rate rises" failed on 5 cases; the MODEL was right (loss-making and non-conventional cash flows). Test restricted, learning recorded
15	2026-10-30	Claude Code	Write edge-case tests (invalid inputs, 100% debt, zero sweep, all cash at booking, loss-making case, late land, long sales window)	tests/test_edge_cases.py	Accept	All identities hold; loss case has zero tax and negative NPV as expected
16	2026-10-30	Claude Code	Test revealed the sensitivity table had 15 columns	AI explained the cause (per-driver step labels) and rewrote the table with positional columns and a Step column	Accept	Visible in screenshot docs/screenshots/03_sensitivity.png
17	2026-10-30	Claude Code	Memo number-check flagged "16.7%" in the template memo	Cushion percentage added to the fact sheet	Accept	check_numbers now returns no warnings for the template memo
18	2026-10-30	Claude Code	Run three more input cases through copies of the Excel workbook (compare_cases.py)	Cash flows matched in all cases. Excel IRR() gave #N/A in the downside case; heavy-debt case gave IRR −13.8% (Python) vs 839% (Excel)	Modify	Excel formula given a 1% guess. Multiple-IRR problem confirmed by locating both roots; tool now detects non-unique IRR and warns. Documented in validation report discrepancy 3 and 4
19	2026-10-30	Claude Code	Independent decimal recomputation of months 1, 2, 8, 12, 20, 30, 31, 36 and NPV from the written assumptions (hand_check.py)	Matches the engine to 1.5e−14	Accept (team to confirm)	Worked example for month 8 reproduced in the validation report; the team's calculator check is still required
20	2026-10-30	Claude Code	Browser-level test of the running app with Playwright and Chrome, with screenshots	24 checks: KPIs, every tab, metric switch, memo, Claude-mode error handling, break-even price gives NPV 0 and IRR 15%, validation error, reset, loss case, horizon extension, recompute time 1.1 s	Modify	Two test failures were test timing (edit immediately after Reset); helper now verifies the value. Screenshot review found inverted tornado colours for cost drivers, fixed
21	2026-10-30	Claude Code	AI's first multiple-IRR detector flagged the base case	Second root found at −98% per year (meaningless). Root count restricted to −10% to +50% per month and an NPV-sign consistency check added	Modify	Base case, no-debt and downside cases: 1 root; heavy-debt case: 2 roots, flagged

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22	2026-10-30	Claude Code	Draft validation report, reflection report and presentation plan	docs/VALIDATION_REPORT.md, REFLECTION_REPORT_DRAFT.md, PRESENTATION_AND_DEMO.md	Modify (team to rewrite)	Reflection individual sections left empty on purpose; team must write in own words and verify every claim
23	2026-10-30	Claude Code	Restart the app server from a script	Command killed itself because the kill pattern matched its own command line; the UI test ran against no server	Reject	Replaced with a kill by port number. Logged as a tooling error
24	2026-10-30	Claude Code	Browser timing check: one input change took 27.7 s	AI traced it to the sensitivity loops calling appraise() 56 times, each running two break-even bisections (300 ms) it did not need	Accept	Break-even made optional inside loops; recompute now 1.7 s. Would have stalled the live demo

| 25 | 2026-11-05 | Claude Code | Add the team's AIML API key as a second memo provider (OpenAI-compatible gateway) alongside Anthropic direct | AI added a provider selector, a model field, key loading from `.streamlit/secrets.toml` (gitignored), and the same fact-sheet-only system prompt for both providers | Accept (team to confirm) | Key verified by listing the gateway's models; key stored only in a gitignored file; nothing else in the model changed | | 26 | 2026-11-05 | Claude Sonnet 5.5 via AIML API | Write the base-case investment memo from the fact sheet | 533-word memo in 7 sections, 12 s. Recommends "Proceed, with caution"; ranks price, construction cost, launch delay, land cost as drivers; notes a 3-month construction delay raises NPV; says the fact sheet does not define the scenarios | Modify | Number check: 0 of about 60 numbers unmatched. Read in full by a person. Memo was right that scenario definitions were missing: fact sheet extended, memo regenerated. Delay observation is correct for this model (no delay penalty) and is kept as a caveat. Saved verbatim in `docs/llm_memo_claude-sonnet-5-5.md` | | 27 | 2026-11-05 | Claude Opus 5.5 via AIML API | Same prompt, same fact sheet, larger model | 557-word memo, 17.5 s, same recommendation and same driver ranking | Accept (as a cross-check) | 0 unmatched numbers. Two models agreeing on the narrative does not verify the numbers; the Excel benchmark does. Saved verbatim in `docs/llm_memo_claude-opus-5-5.md` | | 28 | 2026-11-05 | Claude Code | Browser test failed 3 checks after the memo change | AI first suspected its own change, then measured: the Streamlit "Running" indicator appeared 1.2 s after the keystroke on a loaded machine and the test read the old value. The app was correct | Accept | Test fixed to wait for the indicator to appear before waiting for it to disappear; 27/27 pass. Lesson: a failing test is evidence about the test as often as about the tool |

| 29 | 2026-11-05 | Claude Code with web search | Data collection: find published sources for all 21 base-case inputs for a Mumbai (MMR) project | 19 sources found: Knight Frank H2 2025 (Mumbai avg ₹8,856/sq ft), JLL 2024 land ₹17 cr/acre MMR, MCHI approval-cost paper (23% to 32% of project cost in BMC limits), RBI Oct 2026 repo 5.50%, EY/NSE 2024 real estate cost of equity 16.4%, s.115BAA 25.17%, RERA s.13 10% cap, NBFC construction finance 11% to 14% | Modify | Four primary documents (Knight Frank, EY/NSE, MCHI, RBI resolution) were opened and read; the rest were seen only as search summaries and are marked so in `docs/DATA_SOURCES.md`. Two pages (99acres, Business Standard) blocked automated access. The AI's first RBI figure (5.25%) was stale: the rate was raised to 5.50% on 7 Oct 2026, caught by cross-checking rbi.org.in. CPWD PAR 2025 could not be retrieved; a blog's unverified figure was rejected | | 30 | 2026-11-05 | Claude Code | Decide whether to change the base case to the sourced values (marketing 3% to 4%, instalments 18+ months, discount rate 16.4%) | AI recommended keeping the base case and documenting the gaps, because changing inputs would invalidate the Excel benchmark, validation report and screenshots two days before the viva | Accept (team to confirm) | Gaps are stated in DATA_SOURCES §4 and the manual §10. The sensitivity rows at +1 and +2 pp discount rate and +20% costs show the base case stays viable under the sourced values |

| 31 | 2026-11-05 | Claude Code with web fetch | Open every source that had only been seen as a search summary and confirm the figure used | 17 of 19 sources opened, including the RERA Act itself (s.13(1) and s.4(2)(l)(D) quoted verbatim). Five figures corrected: Panvel land asking range is ₹3.2 to 12 cr/acre (not 9.2); Mumbai standard construction cost ₹3,086 (not 3,150); the DNA brokerage article is from 2018 and the L&T schedule from 2014; the 99acres price ranges were replaced by four pages that could be opened, which add a *registered* Kalyan rate of ₹7,900/sq ft | Modify | Two sources still need a browser (Business Standard 2020; MDPI paper statistics). One search-summary claim was wrong: findcirclerate does not call its figures estimates. A first attempt to fetch the blocked pages with a browser user-agent was refused by the tooling as bot-check circumvention and not retried |

| 32 | 2026-11-05 | Dinesh Sai (manual check, no AI) | Open the two sources the AI could not reach | MDPI paper opened: section 4.7 gives median IRR hurdle 18% (SD 4.85%) and median MDC 20% (SD 1.41%). Business Standard 2020 article is paywalled | Modify | S14 confirmed and its row rewritten to the figures actually seen; the AI's quoted mean (18.17%) and range (10% to 30%) were dropped as unchecked. S11 withdrawn from the source list; the interest-rate row now rests on the RBI release and the Loan Bazaar rate card only |

| 33 | 2026-11-05 | Claude Code | First pass of the course's manual benchmark (months 1, 8, 36, NPV, break-even) ahead of the Validation Engineer's re-performance | Calculator-style recomputation with 4-decimal rounding, no engine calls for intermediates; months 1, 8, 36 reproduce exactly; NPV 7.4514 vs 7.4515 (rounding); break-even prices found by bisection through the workbook's own formulas (25 LibreOffice recalculations, 7 minutes) and written to Summary!B21/B22 | Modify (team to confirm) | This does NOT satisfy the course requirement by itself: it was done by the AI. It gives Surya a line-by-line template and expected values. Any line Surya cannot reproduce by hand is a discrepancy for section 7 |

Entries to add as the team works: hand calculation of months 1, 8 and 36 (Validation Engineer), Goal Seek break-even prices, any formula the team disagrees with, feedback from a practising engineer or lender if obtained. 33 entries so far; the minimum is 20.